

UNIT	Quantitative Portfolio Management Syllabus for Term Test II
4	Portfolio Optimization in Practice Diversification & Optimization Methods: Naive diversification (1/N rule), Scientific diversification Risk-Based Approaches: Measuring risk contributions, Risk budgeting, Simplified risk parity portfolios, Full risk parity portfolios Covariance Estimation (Brief): Shrinkage estimators, EWMA (Exponentially Weighted Moving Average), Rolling covariance windows Comparing optimization & diversification approaches: Empirical comparison
5	Beyond Diversification: Portfolio Insurance & CPPI Limits of Diversification: When diversification fails, Correlation breakdown, Empirical evidence CPPI – Constant Proportion Portfolio Insurance: Concept, floor, cushion, multiplier, CPPI vs buy-and-hold, Drawdown constraints, CPPI Simulation & Strategy Analysis: Simulating CPPI using GBM, Monte Carlo evaluation, Analyzing performance, drawdowns, floor violations, Designing and calibrating CPPI strategies (light, conceptual)
6	Portfolio Revision, Evaluation & Bond Portfolio Strategies Portfolio Revision: Meaning, need, constraints, Revision strategies Portfolio Evaluation: Sharpe Ratio, Treynor Ratio, Jensen's Alpha Foreign Exchange Risk in Global Portfolios: Nature of FX exposure, Currency risk measurement, Hedging basics Bond Portfolio Management Strategies: Duration-based strategies (immunization); Ladder, Barbell, Bullet; Active vs passive bond portfolio management